

STOCK PRICE PREDICTION USING MACHINE LEARNING

1. Abstract

The stock market is a complex, dynamic system, and predicting stock prices is a challenging task due to its volatility and non-linearity. This project leverages machine learning algorithms, specifically Long Short-Term Memory (LSTM) networks, to predict stock price trends. Using historical data, the model aims to forecast future stock prices for a given stock ticker. The model is trained using the data from Yahoo Finance, processed through normalization techniques, and evaluated based on Mean Squared Error (MSE). This report outlines the purpose, methodology, and results of the stock price prediction model built using Python.

2. Purpose

The primary purpose of this project is to develop a machine learning model capable of predicting future stock prices. The model is designed to:

- Assist investors and traders in making informed decisions based on historical price trends.
 - Explore the capabilities of deep learning (specifically LSTM) for time-series forecasting.
 - Evaluate the effectiveness of LSTM networks in predicting the price of stocks based on historical data.
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3. Literature Review

Stock Price Prediction has been a widely studied topic in the field of finance and machine learning. Traditional methods like **statistical modeling (ARIMA, GARCH)** have been used for time-series analysis. However, the advent of deep learning has opened new possibilities, with techniques such as **LSTM** (Long Short-Term Memory) showing great promise.

LSTM networks, introduced by Hochreiter & Schmidhuber in 1997, are a type of **Recurrent Neural Network (RNN)** designed to learn long-term dependencies. Unlike traditional feedforward networks, RNNs and LSTMs are particularly useful in handling sequential data, which makes them ideal for stock market prediction.

Recent studies have explored the use of LSTM for predicting stock prices, including:

- **(Xing et al., 2019)**: Used LSTM for predicting stock market trends with impressive accuracy by utilizing both historical stock prices and other financial indicators.
- **(Fischer & Krauss, 2018)**: Implemented LSTM-based deep learning models to predict stock returns, outperforming traditional statistical methods.

This model uses historical stock data from Yahoo Finance to train the LSTM network, predicting stock prices in the future based on patterns observed in the past.

4. Methodology

The methodology used in this project involves the following steps:

1. **Data Collection:**

- Historical stock data for the chosen ticker symbol is fetched using **Yahoo Finance API** via the **yfinance** library.
- The data consists of stock prices including Open, Close, High, Low, and Volume from 2010 to 2019.

2. **Data Preprocessing:**

- The data is split into two sets: **training (70%)** and **testing (30%)**.
- The stock prices are normalized using the **MinMaxScaler** to scale the data between 0 and 1 for better training of the model.

3. **Model Architecture:**

- The model used is a **Sequential LSTM Network** with:
 - Two **LSTM layers** with 50 and 60 units respectively, for learning the sequential patterns.
 - **Dropout layers** to prevent overfitting.
 - A **Dense layer** as the output layer for predicting the future stock price.
- The model is compiled with the **Adam optimizer** and **Mean Squared Error** as the loss function.

4. **Model Training:**

- The model is trained on the historical stock prices, using 100 past days to predict the next day's closing price.

5. **Model Evaluation:**

- The model's performance is evaluated using **Mean Squared Error (MSE)**.
- Predictions are compared to actual stock prices, and visualizations are generated to show how well the model has learned the trend.

5. Algorithms

The main algorithm used for the stock price prediction in this model is:

- **Long Short-Term Memory (LSTM):** This is a type of **Recurrent Neural Network (RNN)**, capable of learning and retaining long-term dependencies in sequential data. LSTM is specifically well-suited for time-series forecasting tasks like stock price prediction.

In addition to the LSTM, the following supporting techniques are used:

- **Min-Max Scaling:** This is a normalization technique that scales the data between 0 and 1 to improve the performance and training efficiency of the neural network.

- **Dropout Regularization:** Dropout layers are added to the model to prevent overfitting by randomly deactivating some neurons during the training process.
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6. Datasets

The dataset used for training and testing the model comes from **Yahoo Finance**, which provides historical stock prices. The data used in this project contains the following attributes:

- **Date:** The date of the stock price record.
- **Open:** The opening price of the stock.
- **High:** The highest price during the trading day.
- **Low:** The lowest price during the trading day.
- **Close:** The closing price of the stock.
- **Volume:** The number of shares traded.

The dataset spans from **January 1, 2010** to **December 31, 2019**, and stock prices for the chosen company (e.g., Google) are used to predict future prices.

7. Analysis

The model's performance was evaluated using **Mean Squared Error (MSE)** to measure the difference between the predicted and actual stock prices. Visualizations of stock prices, including moving averages (100-day and 200-day), were generated to evaluate how well the model captures price trends.

Key findings:

- The model performs reasonably well in predicting stock price trends based on historical data.
 - The prediction accuracy can vary due to the volatile nature of stock prices, and the model should be regularly retrained with updated data.
 - The use of LSTM provides an advantage over traditional machine learning algorithms like decision trees or regression models, particularly for sequential data.
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8. Conclusions

The project demonstrates the potential of **LSTM networks** for stock price prediction. While it performs well on historical data, it's important to note that stock price prediction is inherently **uncertain** due to market volatility and external factors (e.g., economic events, global news).

Key conclusions:

- LSTM can effectively capture long-term dependencies in stock price data.
- The model can be improved by incorporating more features (like news sentiment, technical indicators, etc.).

- The prediction accuracy might improve with hyperparameter tuning and model optimization.
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9. Future Plans

The following steps can be taken for improving the model and expanding the project:

1. **Hyperparameter Tuning:** Experiment with different LSTM configurations (e.g., the number of units, layers, etc.) and optimization techniques.
 2. **Incorporating More Data:** Add more features such as technical indicators (RSI, MACD) or news sentiment analysis for better predictions.
 3. **Model Comparison:** Compare the LSTM model with other machine learning algorithms such as **GRU**, **ARIMA**, or **Random Forest** to evaluate which performs best.
 4. **Real-Time Prediction:** Integrate real-time stock data and build a system to predict stock prices in real-time.
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Conclusion

By using LSTM-based deep learning for stock price prediction, the model demonstrates the power of machine learning in financial forecasting. However, stock price prediction remains a highly complex and uncertain task, and there's a need for further enhancement in model performance through additional features and tuning.